Clinical Decision Support System for Disease Risk Assessment

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Course: **CSIS 4495 – Applied Research Project**

Section: 03

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1. Introduction

1.1 Executive Summary

The trend in health care systems is a phenomenon that offers customers with the access to their personal medical data in digital formats in the form of PDF laboratory reports, scanned medical summaries, structured health records and medical images. Although there has been increased transparency and engagement with patients, the interpretation of these records poses a major difficulty to non-clinical users. Several reports in the medical field are filled with complicated words, numerical signs, and unarranged data that the patients find difficult to comprehend unless advised by a professional.

Available healthcare analytics and clinical decision support systems are mainly aimed at clinicians and healthcare professionals. These systems tend to be based on structured datasets and exclude much of the patient-accessible information that is stored in unstructured forms, e.g. PDFs and scanned documents. Consequently, patients are likely to become unaware of the early risk factors in their personal medical profiles, which restricts their opportunities to make a proper decision or seek medical consultation in time.

This study suggests the creation of a patient-centered clinical decision support system, which would help people to gain knowledge about the risk indicators of diseases based on their own medical history. The suggested system combines multimodal medical information such as PDF reports, elaborated health information, and chosen medical images in the format of JPG/PNG to deliver interpretable and confidence-driven risk decisions. The system does not either carry out medical diagnosis or substitute professional medical judgment, rather it assists the patient in becoming more aware by showcasing potential risk indicators in a clear and understandable way.

The study will presuppose working with anonymized or patient-supplied data and emphasize the ethical machine learning, interpretability, and usability. This project will show that applying data analytics and machine learning can help to improve patient engagement and make decisions about their health through the increase in patient comprehension of their medical data.

2. Framing the Problem

While healthcare systems have succeeded in providing patients with unprecedented digital access to their medical records, a critical gap remains in the ability of patients to utilize this information for proactive health management.

2.1 Interpretation Barrier: Lab reports use complex words and numbers that regular non-clinical users lack in proficiency find confusing.

2.2 Technology Dependency: Current clinical decision support systems are designed specifically for healthcare professionals, leaving patients without accessible tools to measure their own health data.

2.3 Hidden Information: Significant portions of patient data are stored in unstructured formats—such as scans and PDFs—which are often excluded from existing healthcare analytics.

2.4 Unclear Clinical Intervention: Due to this lack of clarity, patients frequently overlook early disease risk indicators present in their records, preventing them from making informed decisions or seeking timely medical advice

Our proposed Clinical Decision Support System for Disease Risk Assessment designed to mitigate the gap between raw medical data and patient understanding. It transforms complex, often unreadable documents into a clear roadmap for personal health management.

3. Literature Review: Past Work or Research

The previous studies in the healthcare analytics field have proved the usefulness of machine learning methods in assisting clinical decision-making and predicting disease risks. Research papers like Rajkomar et al. (2019) indicate that with the help of supervised learning models that have been trained on electronic health records, it is possible to detect the patterns that are related to the risk of developing chronic diseases with high precision. Much of this work, however, takes the clinician facing systems and is largely dependent on structured datasets to be applicable, and this limits applicability to patient accessible records.

As noted in a study by Shortliffe and Sepulveda (2018), the most effective clinical decision support systems should offer timely and interpretable information and not automated diagnoses. This is in line with the increased focus on explainable artificial intelligence (XAI) within healthcare, in which transparency and trust are necessary to ethically implement. Holzinger et al. (2019) also emphasize that the explainable models are especially emphasized when the system communicates with non-expert users, patients, in particular.

Available literature on the subject of unstructured medical data shows that much of the healthcare data is presented in formats like PDFs, scanned documents and clinical notes. It has been demonstrated by prior work that such documents can be successfully extracted by OCR and rule-based natural language processing to derive its clinically relevant indicators, which can then be used in downstream analytics. Nonetheless, few studies have been done to involve the application of these features extracted in patient-facing decision support tools. The given

project is based on these findings and incorporates unstructured document processing with a structured machine learning model to assist patients in understanding them.

4. Assumptions, Hypotheses and Benefits

These studies presume the availability of patient accessible medical records including laboratory reports and summaries, whereby, the records include enough numeric and textual indicators that help in assessing risks of diseases. It is presumed as well that publicly available datasets or anonymized datasets can reasonably reflect common healthcare reporting formats. Also, the project assumes that patients seek informational support instead of diagnostic conclusions and that explainable risk factors could enhance the level of understanding without leading to excessive anxiety.

The hypothesis of this study is that the medical records that can be accessed by the patient, including laboratory reports and summaries, have enough numerical and textual cues to aid in the assessment of the risk of the disease. It is additionally assumed that publicly available or anonymized dataset is a reasonable representation of typical healthcare reporting forms. Also, the project is based on the assumption that patients request informational backup and not diagnostic conclusions and that clarifiable risk factors can enhance understanding with a patient without creating unnecessary anxiety.

The expected advantages of this study are greater patient access to personal health information, awareness of possible threats of disease, and communication between patients and medical workers. Technically, the project offers a multimodal healthcare analytics framework that can be used on a reuse basis. The study has academic implications to applied knowledge in machine learning ethics, explainable AI, and patient-centered system design.

5. Proposed Solution

5.1 Scope and Deliverables:

5.1.1 Risk Indicator Identification: Applying models/algorithms to find and highlight potential disease risk indicators within those documents.

5.1.2 Multimodal Data Processing: Building a system that can read and "understand" information from different formats like PDF lab reports, structured health records, and medical images in JPG or PNG formats.

5.1.3 Patient-Friendly Translation: Converting complex medical terms and numbers into simple, interpretable language for non-clinical users.

5.1.4 Ethical Machine Learning: Focusing on the use of anonymized data and ethical data analytics practices.

5.1.5 Dashboard for Patient's Medical Data: Provide a summarized dashboard for non-clinical users with transparent, explainable data.

6. Research Design, Methodology, and Justification

This paper is an applied research proposal that adheres to the experimental and comparative methodology. The study aims at designing, developing and testing a patient-centered clinical decision support system that employs machine learning to help in disease risk management through a multimodal medical record.

The proposed study will use a quantitative approach, in which the extracted and structured numerical characteristics of medical records are examined via supervised machine learning methods. The training of various models, comparing them to each other, is used to assess the performance, interpretability, and applicability to patient-facing healthcare analytics.

The practicality of the research project guarantees that the product of the research does not just exist as a theory but also leads to a working software system that illustrates the practicality.

6.1. PDF and Medical Report Processing (Core Component)

Data Collection: Patients commonly receive laboratory results and medical summaries in PDF or scanned formats. Sample datasets will include anonymized or publicly available lab report PDFs and synthetic patient records.

Methodology: Text-based PDFs will be processed using PyPDF2, while scanned documents will undergo Optical Character Recognition (OCR) using Tesseract. Rule-based Natural Language Processing techniques, such as regular expressions, will be applied to extract clinically relevant indicators including glucose levels, cholesterol values, and blood pressure readings. Extracted values will be cleaned, normalized, and converted into structured numerical features suitable for machine learning models.

This component forms the core of the research due to its practicality, feasibility, and strong alignment with patient-accessible healthcare data.

6.2. Structured Health Data Processing

Structured patient-provided data such as age, body mass index (BMI), and basic health attributes will be incorporated into the dataset. These features will be combined with extracted report indicators to improve the robustness of disease risk assessment while maintaining transparency.

6.3. Image-Based Risk Indicator (Optional Extension)

Data Collection: Simple medical images such as chest X-rays will be used from publicly available datasets.

Methodology: A pretrained Convolutional Neural Network (CNN), such as ResNet or MobileNet, will be utilized through transfer learning. The image model will output a binary risk indicator rather than diagnostic conclusions. This output serves as a supplementary signal to enhance multimodal analysis while maintaining ethical and technical feasibility for a solo research project.

6.4. Justification

The research design and methodology used are both explained as the rational decisions in the light of the existing literature and previous coursework about the data analytics, machine learning, and healthcare system.

Past research has shown that machine learning models based on feature, including Logistic Regression and Random Forests, are useful in predicting the risk of diseases using structured health data, and they are also more interpretable than black-boxes (Shortliffe and Sepulveda, 2018; Rajkomar et al., 2019). Explainable analytics also falls into the category of responsible AI in the healthcare field (Holzinger et al., 2019).

OCR and rule-based NLP to extract clinical indicators in PDFs is justified by the literature on healthcare informatics on the relevance of unstructured medical records in decision support systems. Using these methods will allow access to patient-readable data that is typically not used in conventional analytics.

The voluntary application of transfer learning to medical images is explained by the fact that it can reach acceptable performance with small datasets, and it makes fewer complex calculations, which makes it applicable to an applied research project.

Lastly, applied research methodology is consistent with the course acquired knowledge in machine learning, data mining, database systems, and software engineering. It can be used to develop a practical system, compare it with other systems, and conduct an ethical assessment to meet the academic and technical goals of the Applied Research course.

7. Expected Results

The expected outcome of the suggested study is to manage to design and test a patient-centered clinical decision support system successfully and make people learn more about the potential risk factors of the disease on the basis of personal medical history. The system will be used to demonstrate that multimodal medical components (including PDF laboratory

reports, scanned documents, formal health information, and chosen medical images) can be effectively digested and integrated into one and readable risk assessment model.

Technically, the study is expected to prove that the trained supervised machine learning models, such as Logistic Regression and the Random Forest are likely to provide credible results in the assessment of the risks of the disease in the case when the study is carried out on the identified features in both structured and unstructured medical records. The trade-offs between the accuracy of a model and its interpretability will be drew using the comparative analysis, which can justify the applicability of explainable models within the sphere of patient-facing healthcare systems. The importance of features analysis is likely to establish the importance of significant health indicators that introduce the most contribution to a specific risk level that is predictable.

The system is also expected to demonstrate that it is able to derive automated information on documents that are made available to the patient. More complex medical reports can be translated into structured and machine-readable characteristics using more complex procedures of OCR and natural language processing technology, which do not require clinical expertise on the part of the user. This finding is worth the usefulness of unstructured healthcare data in decision support tools in practice.

Usability and ethical perspective: The study will result in the open and non-diagnostic decision support interface that will present risk indicators in a way that is understandable to the patient. Hopefully, the users will be more educated as to the dangers to their health yet they will retain a feeling that the system will never replace the professional medical judgement.

8. Contribution to Practical Applications

The implications of this study on practice are useful in a number of significant ways. To begin with, the project illustrates how applied machine learning and data analytics could help to close the gap between raw medical data and patient understanding. This practice is directly applicable in patient portals, personal health record systems, and digital health platforms that strive to enhance the engagement of patients.

Second, the system offers a platform that is scalable on how to process heterogeneous formats of healthcare data that are usually found in real-life situations. The proposed architecture allows healthcare technology developers and researchers to expand the architecture to accommodate more diseases, data sources or populations without necessarily redesigning that system.

Third, the study will add to the existing body of research on patient-centered healthcare analytics by moving the decision support process out of clinician-only contexts and into the patient-friendly platforms that are ethically crafted. With its focus on explainability and transparency, the findings facilitate the practice of responsible AI and reduce the chances of abuse or misunderstanding.

Lastly, the project is an example of how non-diagnostic clinical decision support systems can be used as proof-of-concept to engage the people in a more proactive manner with their

health data, which may prompt them to consult healthcare providers earlier and have healthier interactions with them.

9. Technologies Used

• Operating System / Platform: Windows or Linux

• Programming Language: Python

• Libraries: pandas, NumPy, scikit-learn

• PDF Processing: PyPDF2

• Machine Learning Models: Logistic Regression, Random Forest

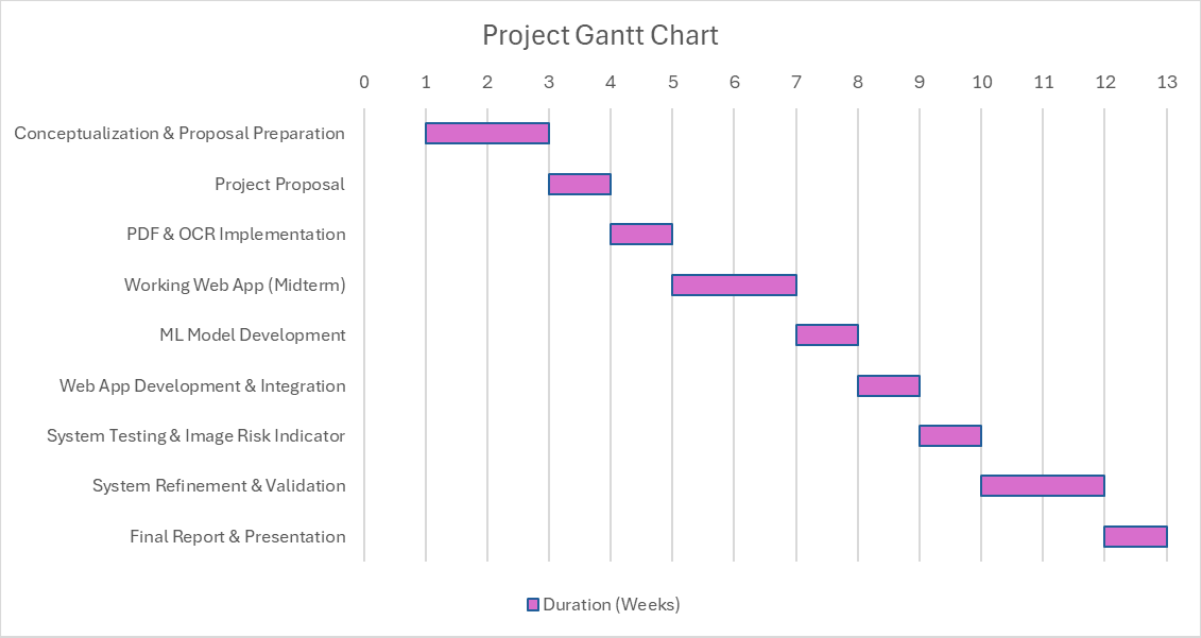
• Deep Learning: TensorFlow or PyTorch (pretrained CNNs)

• Database: PostgreSQL or SQLite

• Backend: Flask or FastAPI

• Frontend: Streamlit (patient-facing interface)

10. Project Planning and Timeline



DETAILED IMPLEMENTATION OF GANTT CHART

WEEKS 1–3: PROJECT PLANNING & REQUIREMENTS DEFINITION

User Stories

- As a project team, we need to clearly define the problem scope, system objectives, and technical stack so that development starts with clarity and a shared vision.
- As a student research team, we need to document requirements and timelines to meet course deliverables.

Tasks

- Finalize project topic, scope, and ethical boundaries (non-diagnostic, patient-centered).
- Conduct literature review on clinical decision support systems, multimodal data processing, and explainable machine learning.
- Define system requirements and high-level architecture.
- Select technologies for frontend, backend, database, OCR, and machine learning.
- Create the project proposal, work logs, and project contract.
- Set up the GitHub repository with appropriate folder structure and initial commits.

Deliverables
Project proposal draft, requirements documentation, project contract, work logs, GitHub repository setup.

WEEKS 3–4: PROJECT PROPOSAL

User Story: As a project team, we need to finalize and submit a well-documented proposal that clearly explains the problem, methodology, and expected outcomes.

Tasks

- Refine and finalize all sections of the project proposal.
- Ensure alignment with course requirements and feedback.
- Complete proposal submission with supporting documentation and repository updates.

Deliverables
Approved project proposal, finalized work logs, confirmed repository setup.

WEEKS 4–5: PDF & OCR IMPLEMENTATION

User Stories

- As a patient, I want to upload my medical reports securely so that relevant information can be extracted.
- As the system, I need to convert uploaded documents into usable text data.

Tasks

- Implement secure PDF upload functionality.
- Integrate OCR tools (e.g., Tesseract) for text extraction from scanned medical reports.
- Preprocess extracted text for further analysis.
- Define initial data structures for storing extracted values.

Deliverables
PDF upload module, OCR text extraction pipeline, initial structured data output.

WEEKS 5–7: WORKING WEB APPLICATION (MIDTERM IMPLEMENTATION)

User Stories

- As a patient, I want to interact with a basic web application to submit my data.
- As the system, I need to process uploaded reports and structured inputs end-to-end.

Tasks

- Develop a functional patient-facing web interface.
- Integrate PDF/OCR outputs into the backend pipeline.
- Generate structured features from extracted text (e.g., glucose, cholesterol).
- Document system progress for the midterm report.

Deliverables
Working web application, structured feature generation pipeline, midterm report.

WEEKS 7–8: MACHINE LEARNING MODEL DEVELOPMENT

User Stories

- As a patient, I want to receive an understandable risk assessment based on my data.
- As the system, I need trained models to generate reliable risk indicators.

Tasks

- Implement machine learning models such as Logistic Regression and Random Forest.
- Train models using structured feature data.
- Evaluate models using appropriate performance metrics.
- Select the best-performing model for deployment.

Deliverables
Trained ML models, evaluation results, documented model selection.

WEEKS 8–9: WEB APPLICATION INTEGRATION

User Stories

- As a patient, I want to see my risk assessment results clearly displayed in the application.
- As the system, I need to connect model predictions with the user interface.

Tasks

- Integrate ML model predictions into backend services.
- Connect frontend components to prediction APIs.

- Display risk level, confidence score, and contributing indicators in the UI.

Deliverables

Integrated web application with ML-based risk assessment outputs.

WEEKS 9–10: SYSTEM TESTING & IMAGE-BASED RISK INDICATOR

User Stories

- As a patient, I want the system to behave reliably across different inputs.
- As the system, I need to be tested for correctness and robustness.

Tasks

- Perform end-to-end system testing.
- Validate data flow and output correctness.
- (Optional) Integrate a pretrained CNN for a simple image-based risk indicator.

Deliverables

Tested system workflow, optional image-based risk indicator integration.

WEEKS 10–12: SYSTEM REFINEMENT & VALIDATION

User Stories

- As a patient, I want outputs that are easy to understand and not overwhelming.
- As the project team, we want a stable and polished system for final submission.

Tasks

- Refine UI/UX for clarity, accessibility, and usability.
- Improve error handling and edge-case management.
- Conduct usability testing and validate system outputs.
- Finalize features within the approved project scope.

Deliverables

Refined and validated web application, usability improvements, finalized feature set.

WEEKS 12–13: FINAL REPORT & PRESENTATION

User Stories

- As a project team, we need to clearly communicate our work, findings, and contributions.
- As evaluators, instructors need complete documentation and a working demonstration.

Tasks

- Complete the final project report, including methodology, implementation, results, and discussion.
- Prepare final presentation slides and system demonstration.

- Update GitHub repository with final code and documentation.
- Prepare for final presentation and oral defense.

Deliverables

Final project report, presentation slides, working web app demo, updated GitHub repository.

11. Project Contract

Project Contract Agreement

We, the undersigned students, agree to complete the Applied Research Project titled:

"Clinical Decision Support System for Disease Risk Assessment"

in accordance with the scope, objectives, methodologies, and timeline specified in this proposal.

Each team member agrees to:

- Fulfill their assigned responsibilities
- Maintain accurate and frequent work date/hour logs
- Adhere to the proposed milestones and deliverables
- Communicate any scope or timeline changes in advance
- Follow academic integrity and ethical research guidelines

Team Responsibilities

- **Uthara Uday:** Project coordination, system architecture, machine learning modeling, documentation
- **Akiful Mohaimin:** Data extraction research, OCR pipeline support, web application development, testing

Meeting Format

- Weekly team check-ins (online and offline)
- Ad-hoc meetings as required during major milestones

Project Duration

- January 2026 – April 2026

Team Lead: Akiful Mohaimin

Name	**Signature**	**Date**
Uthara Uday	Uthara Uday	26th Jan 2026

Akiful Mohaimin	Akiful Mohaimin	26th Jan 2026

12. AI Use Section

AI Tool Name	Version & Account Type	Specific Feature for Which the AI Tool Was Used	Value Added by Student
ChatGPT	GPT-5.2, Premium	Guidance on creating a Gantt chart in Excel and structuring a week-based project timeline (Prompt 1).	Adapted the guidance to the course schedule, validated week overlaps, and independently implemented the final Gantt chart.
ChatGPT	GPT-5.2, Premium	Conceptual clarification of what a Clinical Decision Support System (CDSS) is and how it differs from automated medical diagnosis (Prompt 2).	Applied the explanation to ensure ethical, non-diagnostic, and patient-centered framing throughout the project.
ChatGPT	GPT-5.2, Premium	Explanation of OCR-based data extraction from PDF medical reports for machine learning applications (Prompt 3).	Translated the conceptual explanation into project-specific system architecture and feature extraction design.
ChatGPT	GPT-5.2, Premium	Identification of suitable machine learning algorithms for disease risk prediction using structured healthcare data (Prompt 4).	Selected appropriate algorithms aligned with project scope and documented their intended use.
ChatGPT	GPT-5.2, Premium	Discussion of ethical considerations for a patient-facing machine learning system analyzing medical records (Prompt 5).	Incorporated ethical considerations into the problem statement, scope, and system limitations.
ChatGPT	GPT-5.2, Premium	Verification of alignment between the final project document and the professor's requirements (Prompt 6).	Reviewed, revised, and finalized the document to ensure compliance, clarity, and academic integrity.

13. Work Date / Hours Log

Student Name: Uthara Uday

Date	Number of Hours	Description of Work Done
Jan 13, 2026	1.0	Initial brainstorming on automated scheduling system as potential Applied Research Project topic
Jan 15, 2026	1.0	Evaluated feasibility and scope of automated scheduling system against CSIS 4495 requirements
Jan 16, 2026	1.5	Team discussion with Akiful on project ideas and comparison of proposed topics
Jan 17, 2026	1.0	Reviewed Akiful's proposed clinical decision support system idea
Jan 18, 2026	1.5	Contributed to refining scope and ethical framing of clinical decision support system
Jan 19, 2026	1.0	Assisted in defining patient-centered focus and non-diagnostic constraints
Jan 24, 2026	1.5	Co-authored proposal sections: Introduction and Research Objectives
Jan 26, 2026	1.0	Reviewed and refined methodology and system architecture sections

Student Name: Akiful Mohaimin

Date	Number of Hours	Description of Work Done
Jan 13, 2026	1.5	Initial ideation of patient-centered clinical decision support system

Jan 15, 2026	1.0	Conducted preliminary research on clinical decision support systems
Jan 16, 2026	1.5	Conducted preliminary research on clinical decision support systems
Jan 17, 2026	1.0	Discussed project scope and feasibility with team member
Jan 18, 2026	1.5	Led refinement of project focus and ethical constraints
Jan 19, 2026	1.0	Designed high-level multimodal system workflow
Jan 25, 2026	1.5	Drafted methodology and data analysis approach
Jan 26, 2026	1.0	Reviewed full proposal for technical accuracy and clarity

14. Closing and References

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15. APPENDIX

1. how to create gantt chart in excel?
2. Explain what a Clinical Decision Support System (CDSS) is and how it differs from an automated medical diagnosis system?
3. Explain how OCR can be used to extract information from PDF medical reports for machine learning applications.
4. What machine learning algorithms are suitable for disease risk prediction using structured healthcare data?
5. Suggest ethical considerations for a patient-facing machine learning system that analyzes medical records.